

NHS-EAD Forecast Submission Report

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Task

We forecast estimated avoidable emergency-department deaths for the 173 rolling 10-day windows of the assessment period. The deliverable is `submission/pred_matrix.csv`: one row per forecast window, columns `forecast_id`, `day_1`, ..., `day_10`.

Each origin uses only information available at that origin, and the target series is treated with the contest's three-day reporting lag, so at origin D the most recent observed target value is from D-3.

Data and leakage discipline

The pipeline uses only the provided BNSSG operational series; no external data is introduced. Same-day operational covariates respect the midday cut-off (values timestamped after midday on the origin day are assigned to the next day). Imputation and rolling summaries are computed per origin from past values only, so no future information leaks into a forecast. The raw challenge dataset is not bundled in this repository.

Model

The final forecast uses a fixed split by horizon:

- `day_1` to `day_5`: Chronos + NB-INGARCH-AQ + seasonal residual correction.
- `day_6` to `day_10`: NB-INGARCH-AQ.

NB-INGARCH is a negative-binomial integer-valued GARCH model: a count time-series process that is self-exciting across days and absorbs the overdispersion of the boarding series. The AQ ("adaptive quantile") variant shifts toward a higher predictive quantile when a small set of operational covariates, known at the origin, indicate elevated pressure; this responds to the asymmetric cost of under-predicting winter spikes under squared error. Chronos is a pretrained, open-source time-series foundation model applied zero-shot to the target history, used only in the shorter-horizon block. The seasonal residual correction is calendar-driven and applied to the `day_1` to `day_5` block.

The horizon split is fixed before the assessment run and does not change from one assessment origin to another except through the horizon number. The committed JSON artifacts record the validation and evaluation quantities used to freeze the candidate.

Validation

We tuned the small number of free parameters (the blend weights and the seasonal-prior scale) on a held-out validation block carved from the pre-assessment period, then evaluated on the combined winter block (Test-A plus Test-B), the closest available mirror of the October-March assessment

season. Significance was assessed with a moving-block bootstrap over forecast origins. The selected per-prize configuration:

Horizon block	Model	A+B MSE
days 1-5	Chronos + NB-INGARCH-AQ + seasonal residual correction	0.1157
days 6-10	NB-INGARCH-AQ	0.1293
combined	fixed horizon split	0.2450

The comparison set also included plain Chronos, NB-INGARCH without AQ, NB-INGARCH-AQ alone, blended Chronos/NB variants, and gradient-boosted and other tree learners. The tree learners generalised poorly across seasons and were not selected. Short and long horizons are dominated by different signals, which is why the per-prize split outperforms any single model across both prizes.

Runtime

The pipeline runs on CPU. One 10-day forecast completes well within the one-hour limit on a standard desktop; the foundation-model component is small and does not require a GPU.

Reproducibility

The repository contains the model-development code, configuration, frozen evaluation artifacts, this report, the forecast matrix, and the validation script. The submitted matrix has 173 rows, the template columns, sequential `forecast_id` values from 1 to 173, and finite numeric forecasts.

`submission/pred_matrix.csv` SHA256:

`513dcaa45cb761c25b43c24c3835e9cb5212d21e68b57f11512367d06be8de3b`

Limitations

The assessment winter included supply-side shocks (a seasonal influenza surge and periods of industrial action) that the provided demand-side covariates capture only partially. The adaptive-quantile gate is the main mechanism that lifts forecasts when system pressure is elevated. The seasonal-residual component is calendar-driven and is applied only to the short-horizon block.